

Selecting the right SoC

An IoT system needs to sense, communicate, perform inference and computation, while taking the security of the system and data into consideration. These things require tight integration of hardware-software architecture. Designers need to design compact products, running on minimum power, and utilising really low bandwidth while costing extremely low.

To achieve all of this the selection of an optimum system-on-chip (SoC) is very critical. SoC is one of the main components that play a crucial role in integrating all the components in an embedded system.

In addition to the discussed problems, the SoCs must last for years and, in many cases, last over a decade on a single coin-cell battery. So, very tight integration of all the aspects is necessary for a good IoT solution. To overcome the problem, engineers are trying to simplify inferencing, and to achieve that we can observe the development of Micro Edge.

Inference at Micro Edge

The inference is turning raw data from the sensors and other IoT devices into a piece of actionable information. In the inference we have seen a development of Micro Edge. Although sounding too fancy, Micro Edge is nothing but where your sensors and devices are residing.

According to Manish, “When you do inferencing right at the sensor or near the device, you are taking care of security, you are sending that inferred solution or the inference across as opposed to the live data. So, it gives you lower latency. It gives you a more secure solution, and obviously, it is cost-effective.” He adds, “This is a growing use-case where inferencing is occurring right at what we call Micro Edge by the sensors and devices out there in the field.”

Need for configurable system

The IoT applications are very diverse.

From a light bulb to a health monitor, to machinery in a factory to an asset tracker, we have diverse applications and various micro verticals, with very different requirements for each one of them in every vector.

Therefore, designing a one-fit solution for all situations is not possible. But being a low-cost solution, it is also not feasible for a company to do R&D from ground zero for every single product. You must do quick turn derivatives in order to stay relevant.

**SOC SENSES
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Furthermore, there are industrial and commercial customers with massive quantity requirements who want to have their own ecosystems and not of Google, Amazon, etc for privacy, security, and confidentiality. So, new players in the IoT segment have a wide variety of opportunities to tap in.

“We have diverse applications and various micro verticals, with different requirements for each one of them in every vector, whether it is sense requirements, compute requirements, etc. Your hardware-software needs to be configurable,” adds Manish.

Finding the right opportunity

IoT has the potential to change how we do a lot of things. We are not just connecting things, or inferring things because it's cool, but because it has the potential to change the world. Apart from enhancing convenience and improving the lifestyle, Manish also

feels that it will transform a variety of industries; it will transform even those industries that are considered low-tech, such as the construction industry.

Manish gives an example of how IoT can become beneficial for the construction industry, which does not usually require automation. He claims that the maximum amount of time and resources are not efficiently managed in the construction industry, and deadlines often get extended due to multiple reasons, such as improper management of inventory, work getting delayed, or resources being wasted due to bad weather or improper working conditions.

These are the areas where IoT can streamline things. IoT can help get live data of these tools, machines, and equipment, the utilisation of the equipment and tools, and their efficiencies. All these parameters are very critical for delivering projects on time.

Manish also believes that there are other parameters, such as how and when you service and maintain your equipment, or the status of the secure access for some of these construction sites where there is a danger of operation, like in mining operations, where you have to take personal safety into account. So, how do you ensure secure access?

There are clear problems right in front of us that we need to identify and solve to improve lives, improve businesses, their efficiencies, and the money that they make. This is just one industry. Nobody's going to give us a lot of money to just look cool. And unless we start solving real problems using IoT, the technology will not take off.

What the future holds

“IoT will disrupt multiple industries in the coming decade and beyond.”

Disruption occurs due to technologies that are generally cheaper and simpler to use. Initially, in terms of performance, these disruptive